

DuitNow QR Merchant Switch Assessment Brief

Organization: Synthetic PayNet Retail Rail Lab

Scenario: DuitNow QR Merchant Switch

Classification: synthetic test evidence only.

Public inspiration: DuitNow QR is publicly described as a single QR payment experience across participating banks and e-wallets.

System summary: Synthetic national QR merchant payment switch for wallets, banks, acquirers, merchant apps, and near-real-time notifications.

Risk goal: High-volume retail edge with mobile wallet and acquiring-bank exposure.

Key findings: TLS 1.2 ECDSA-P256 on merchant callback gateway; ECDH wallet-to-switch channel; RSA-2048 QR payload signing compatibility path; AES-256 settlement vault as low-risk control.

Expected scan behavior: Internet-edge and partner-facing findings should rank high where TLS 1.2, ECDSA, ECDH, and RSA signing appear near QR payment ingress and merchant notification APIs.

Recommended validation: confirm whether merchant callbacks and participant channels can enforce TLS 1.3 hybrid ML-KEM and ML-DSA signatures.

No credentials, private keys, payment tokens, PINs, account numbers, customer data, or live infrastructure references are included.